

ANISH BOKARE

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EDUCATION

VIT Bhopal University

Integrated M.Tech in Computer Science (Cybersecurity); CGPA: 8.20 2023 – 2028

School of Scholars

Higher Secondary Education; 85% (11th), 76% (12th) 2021 – 2023

TECHNICAL SKILLS

- **ML, DL & Federated Learning:** Supervised & Unsupervised Learning, Deep Learning, CNNs, RNNs, Transformers, NLP, Computer Vision, Anomaly Detection, Federated Learning (FL), FedAvg, Differential Privacy, Flower, TensorFlow Federated, Distributed Training
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV, XGBoost, MXNet, Caffe, Pandas, NumPy
- **Data & MLOps:** Data Manipulation/Transformation, Model Selection, Model Training, Cross-Validation, Deployment at Scale, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Spark, MLflow, Docker, Kubernetes, CI/CD, Model Monitoring
- **Statistical & Cloud Tools:** R, SAS, MATLAB, SQL, AWS SageMaker, Azure ML, GCP Vertex AI, Git, Jupyter, VS Code, Linux

WORK EXPERIENCE

MP Police (Cyber Cell)

CyberSecurity-ML Intern Hybrid Sep 2024 – Oct 2025

- Built ML models for anomaly detection and log analysis, improving accuracy by 30% and reducing false positives by 40% via feature engineering, cross-validation, and hyperparameter tuning.
- Developed Python data pipelines (Pandas, NumPy, scikit-learn, TensorFlow) for preprocessing, feature extraction, and training, reducing manual prep time by 35% and enabling deployment at scale.
- Implemented a federated learning prototype with Flower and PyTorch for privacy-preserving threat intelligence across distributed nodes, enabling secure model aggregation without centralizing sensitive data.

COEP Technological University

Research Intern On-Site May 2025 – Jul 2025

- Researched deep learning and federated learning for privacy-preserving data analysis, benchmarking CNN and transformer architectures on structured and unstructured datasets.
- Analyzed complex datasets and improved model performance through feature selection, cross-validation, and hyperparameter optimization.
- Partnered with cross-functional teams to document findings and create knowledge-base articles/SOPs for reproducible ML experiments.

PROJECTS

Federated Learning for Collaborative Medical Imaging Diagnosis

- Architected a privacy-preserving federated learning system using Flower and PyTorch to train a pneumonia detection model across 5 hospitals without sharing patient data; implemented secure aggregation and differential privacy (Opacus).
- Achieved 96% diagnostic accuracy (12% over local models); deployed on AWS with Kubernetes, Terraform, and a real-time monitoring dashboard.

Real-Time Fraud Detection with Streaming ML and Concept Drift Adaptation

- Built a streaming ML pipeline with Kafka, Flink, and PyTorch processing 10,000+ events/sec at sub-50ms latency; integrated ADWIN drift detection, reducing false negatives by 25%.
- Deployed on Kubernetes with auto-scaling and MLflow for versioning/rollback, maintaining 99.9% uptime with seamless model updates.

LLM-Powered Code Review and Vulnerability Detection with RAG

- Fine-tuned CodeLlama-7B with LoRA and RAG using a FAISS vector database of secure coding guidelines, achieving 92% precision and 88% recall on OWASP Top 10 across 10,000+ snippets.
- Integrated as a GitHub Action with inline PR comments and deployed on AWS SageMaker with autoscaling, reducing manual review time by 50%.

ACHIEVEMENTS

SIH India 2023 – College Finalist. | NASSCOM Agentic AI Hackathon – Global Finalist | SolVIT Hackathon 2025 – 3rd Position (Amongst 150+ teams)

EXTRA-CURRICULAR ACTIVITIES

Android Club, VIT Bhopal | R&D Co-Lead

Apr 2024 – Jul 2025

- Led a team of 15+ in ML and AI-focused R&D projects. Facilitated collaboration between development and data science teams to drive the implementation of ML feature enhancements and model deployment.